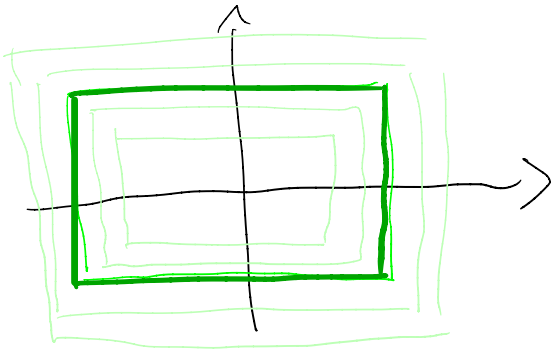
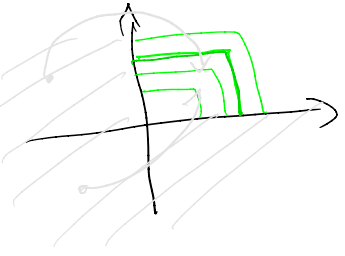
We want



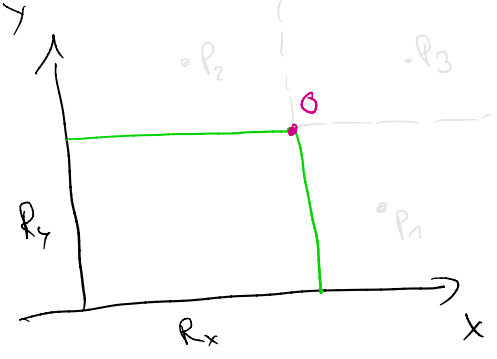
For point P

1. outside: +, value increasing
2. on boarder: 0
3. inside: −, value decreasing

1. We only work in one quadrant with abs(P)



2. Case outside / on boarder



P_1: $d = P_{1x} - R_x$

P_2: $d = P_{2y} - R_y$

P_3: $\sqrt{(P_{3x} - R_x)^2 + (P_{3y} - R_y)^2}$

→ we could implement this with if statements, e.g. if $P_x > R_x$ & $P_y < R_y$

But we can also be smart about it ...

3. The P_1, P_2 computations are part of the computation for P_3 and we can make use of that!

→ If we want to compute P_1, meaning x, the the P_2 part, meaning for y, should become 0 and vice versa

E.g. $\sqrt{(P_x - R_x)^2 + (P_y - R_y)^2}$

this should become zero $\sqrt{(P_x - R_x)^2} = P_x - R_x$

→ We can do this by setting the components to zero, depending on their sign!

E.g. if we want P_1, then P_2 is alway negative and vice versa

→ Hence, we need to take the $\max(\dots, 0)$ for each term

↳ detects negativity and sets then the whole term to zero

$$d = \sqrt{\max(P_x - R_x)^2, 0) + \max(P_y - R_y)^2, 0)}$$

4. in vector notation

e.g. $P = (x, y)$ $R = (S_x, S_y)$

base distance

$d = \text{length}(\max((\text{abs}(P) - R), 0))$

$\sqrt{()^2}$ < 0 map to 1. quadrant

5. Case inside

→ First the overall switch to this case with

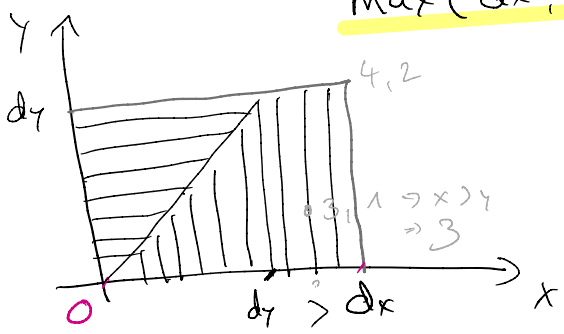
$\ldots + \min(\ldots, 0)$

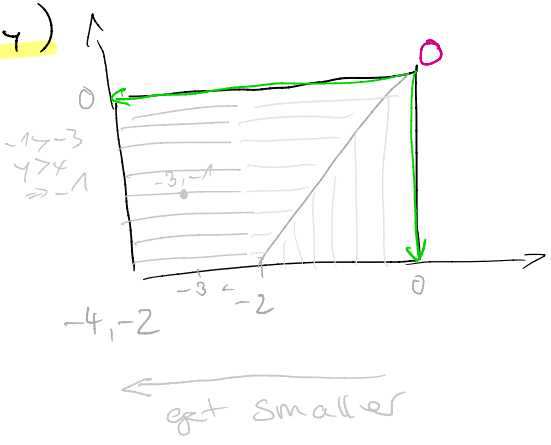
becomes zero when negative

becomes zero when positive hence we only keep it when negative

→ First explanation in a slightly modified scenario to make it easier

$\max(d_x, d_y)$





get smaller

$\rightarrow \quad \ldots + \min(\max(d_x, d_y), 0)$